RiceNet: A Multimodal CNN–Vision Language Framework for Explainable Rice Leaf Disease Classification

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Abstract

Rice leaf diseases greatly reduce crop yields and threaten food security, particularly in South Asia where rice is a key staple. Manual diagnosis is time-consuming, subjective, and often unreliable under field conditions. Although deep learning has shown promise for plant disease classification, conventional CNN-based methods often lack semantic understanding and may struggle with class imbalance and limited generalization. To address these challenges, this study introduces RiceNet, a multimodal CNN–Vision Language Model (VLM) framework for explainable rice leaf disease classification. The architecture combines an EfficientNet-B4 backbone with a BLIP vision encoder through bidirectional cross-attention to jointly learn spatial and semantic features, while an adaptive gated fusion mechanism integrates the multimodal representations. A hybrid Focal Loss–PolyLoss is employed to improve minority-class recognition, while Sharpness-Aware Minimization (SAM), Stochastic Weight Averaging (SWA), progressive resizing, MixUp, and CutMix are incorporated to improve optimization stability and generalization. Evaluated on an eight-class rice leaf disease dataset containing 3,646 training, 770 validation, and 772 test images, RiceNet achieved 96.1% accuracy. Statistical significance analysis using McNemar's test showed significant differences between RiceNet and both the CNN+Transformer ($p < 0.0001$) and CNN+VLM (Frozen) ($p = 0.0002$) baselines, with odds ratios of 8.21 and 4.12, respectively. Grad-CAM visualizations and BLIP-generated captions further provide complementary visual and semantic explanations. Overall, the results demonstrate that RiceNet provides accurate, statistically supported, and interpretable rice leaf disease classification, highlighting its potential for practical precision agriculture applications.

*Both authors contributed equally to this research.

CCS Concepts

• **Computing methodologies → Computer vision**; **Neural networks**; **Machine learning**; • **Applied computing → Agriculture**.

Keywords

Rice leaf disease classification; multimodal learning; vision-language model; cross-attention; precision agriculture.

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1 Introduction

Rice (Oryza sativa) is a crucial staple crop globally, providing the primary food source for over half of the world's population. In South Asia, it accounts for nearly 70% of daily calories, essential for food security, economic stability, and sustainable agriculture [9]. However, rice is highly susceptible to various leaf diseases caused by bacteria and fungi, such as Bacterial Leaf Blight, Brown Spot, Leaf Blast, Leaf Scald, Narrow Brown Leaf Spot, Rice Hispa, and Sheath Blight [14]. These diseases decrease crop yields and grain quality, leading to significant economic losses for farmers and threatening food supply chains in developing nations. Often, delayed diagnosis and improper treatment can cause yield reductions of 30% to 50%

under adverse environmental conditions [3]. Consequently, precise and prompt disease detection is vital for precision agriculture and sustainable crop management. Traditionally, diagnosing rice leaf diseases relies on manual visual inspections by agricultural experts or plant pathologists. While this method has been used for decades, it faces significant challenges. Manual inspection is labor-intensive, time-consuming, subjective, and heavily dependent on expert knowledge. Additionally, rural areas often lack access to trained specialists, making early diagnosis difficult. Variations in environmental conditions-such as lighting, leaf orientation, complex backgrounds, and overlapping symptoms-further hinder visual identification. As a result, there is a growing need for automated, scalable, and intelligent disease detection systems that can support farmers in real-field conditions.

Recent advances in deep learning have significantly improved plant disease diagnosis. CNNs and ImageNet-pretrained models such as VGG16, AlexNet, and ResNet50 have achieved strong results in image-based disease classification by automatically learning disease-related features [8, 18]. However, their performance can be limited by overfitting on small or imbalanced agricultural datasets and poor generalization under variations in leaf appearance, lighting, and disease severity [21]. The EfficientNet family [24] addresses these limitations through compound scaling of network depth, width, and input resolution while maintaining computational efficiency. EfficientNet variants have achieved 94–97% accuracy on rice leaf disease datasets [11, 12], while EfficientNet-B4 provides greater representational capacity for fine-grained disease symptoms [13]. Nevertheless, these CNN-based approaches remain unimodal, relying primarily on visual information and lacking semantic and contextual disease knowledge, which can limit their ability to distinguish visually similar diseases. To enhance feature representation and target disease-relevant regions, attention mechanisms have been integrated into CNN-based disease classification systems. A commonly used module is the Convolutional Block Attention Module (CBAM) [10], which sequentially applies channel and spatial attention to highlight important areas and suppress background noise. CBAM-enhanced CNN models have demonstrated better performance in detecting rice bacterial leaf blight and in other agricultural image analysis tasks [5]. Although attention mechanisms boost the learning of discriminative features, they still operate solely within the visual domain and do not provide semantic insights into disease traits. Consequently, these methods may struggle with ambiguous or visually similar disease patterns.

Recently, Vision-Language Models (VLMs) have become a key focus in multimodal artificial intelligence. For example, BLIP (Bootstrapping Language-Image Pre-training) [4] can learn combined visual and textual representations through extensive multimodal pretraining. Originally designed for image captioning and vision-language tasks, BLIP extracts rich semantic features that enhance traditional CNN outputs [7]. Cross-modal fusion methods, which combine visual and semantic information, have shown significant improvements across various fields, including medical imaging, scene analysis, and crop disease detection [22]. By merging image features with semantic context, VLM approaches can better identify disease-specific traits and remain robust under difficult conditions. Nonetheless, research on combining CNN architectures with BLIP vision encoders via bidirectional cross-attention is still limited, especially for rice leaf disease classification. Class imbalance is another significant challenge in agricultural disease datasets. In real-world settings, some diseases are much rarer than others, causing highly skewed class distributions. Traditional loss functions like categorical cross-entropy tend to favor majority classes, leading to poor performance on minority classes. Focal Loss [26] mitigates this problem by down-weighting the easier samples and focusing on harder or misclassified ones. It has proven effective in object detection and plant disease classification [25]. PolyLoss [20], which modifies classification loss functions via polynomial expansion, further enhances optimization and accuracy in imbalanced learning scenarios. Recent research in medical image analysis indicates that combining Focal Loss and PolyLoss can boost both robustness and minority-class detection [1]. However, the use of such hybrid loss strategies in rice leaf disease classification remains underexplored.

Besides architecture and loss design, optimization strategies are essential for enhancing generalization performance. Standard stochastic gradient descent (SGD) often converges to sharp minima, which may reduce training error but result in poor generalization to new field conditions. Sharpness-Aware Minimization (SAM) [2] tackles this issue by explicitly seeking flatter minima in the loss landscape, thus boosting robustness and generalization. Similarly, Stochastic Weight Averaging (SWA) [16] enhances training stability by averaging weights over multiple epochs, leading to smoother optimization paths and broader optima. Progressive resizing, where image resolution gradually increases during training, also boosts convergence efficiency and final accuracy [23]. While these techniques have shown strong results individually in computer vision, few studies have explored combining SAM, SWA, and progressive resizing in a single framework for rice leaf disease classification. Data augmentation methods are vital for boosting robustness and minimizing overfitting in deep learning models used in agriculture. Techniques like MixUp [26] and CutMix [25] create synthetic training data by merging images and labels, which enhances model regularization and smooths decision boundaries. Ensemble learning and test-time augmentation (TTA) strategies further improve the robustness and stability of classifications [1]. Additionally, explainable artificial intelligence (XAI) methods are gaining importance for real-world agricultural applications. Grad-CAM [20] is a popular visualization tool that highlights disease-related areas in plant images. Nonetheless, most current methods only produce visual heatmaps without accompanying textual explanations, which can limit interpretability and user confidence.

Despite notable progress in earlier research, several key issues still need to be addressed. Firstly, most current methods depend only on unimodal CNN architectures and lack semantic contextual understanding. Secondly, traditional loss functions are not well-suited to handle the severe class imbalance common in agricultural datasets. Thirdly, standard optimization techniques tend to converge at sharp minima, which can limit their ability to generalize in real-world scenarios. Lastly, many existing systems offer limited interpretability and do not provide multimodal explanations that integrate visual and semantic reasoning. To overcome these challenges, this study introduces RiceNet, a hybrid CNN–Vision Language Model framework for rice leaf disease classification. The main contributions of this work are summarized as follows:

- A multimodal architecture is developed by integrating an EfficientNet-B4 backbone with a BLIP vision encoder to jointly learn spatial and semantic disease representations.
- A bidirectional cross-attention mechanism is introduced to effectively fuse CNN-based visual features with vision-language semantic features, improving discrimination between visually similar disease classes.
- A hybrid loss function combining Focal Loss and PolyLoss is employed to address class imbalance and improve minority-class recognition performance.
- Advanced optimization and augmentation strategies, including Sharpness-Aware Minimization (SAM), Stochastic Weight Averaging (SWA), progressive resizing, MixUp, and CutMix, are incorporated to enhance model robustness and generalization capability.
- Grad-CAM visualizations and BLIP-generated textual descriptions are integrated to provide multimodal interpretability and improve model transparency.

Experimental results on an 8-class rice leaf disease dataset demonstrate that RiceNet outperforms previous baseline versions and conventional CNN-based approaches. The findings indicate that combining multimodal semantic understanding with optimization-aware training significantly improves disease classification performance, particularly for minority and visually similar disease categories. Overall, this work contributes toward developing reliable, explainable, and robust deep learning systems for practical precision agriculture applications.

2 Methodology

This section outlines the full methodology of the proposed multimodal rice leaf disease classification framework. It combines CNNs, vision-language modeling, bidirectional cross-attention, imbalance-aware learning, and advanced optimization into a unified architecture to enhance accuracy, generalization, and interpretability in real agricultural settings. Unlike traditional systems relying solely on visual data, this framework integrates spatial and semantic features to better differentiate disease categories. The architecture is shown in Figure 2, with the training process in Figure 1. The methodology involves dataset preparation, preprocessing, augmentation, feature extraction, cross-modal interaction, feature fusion, classification, optimization, and evaluation. Each stage addresses limitations of existing rice disease methods, such as class imbalance, semantic gaps, poor generalization, and limited interpretability.

2.1 Dataset Description and Problem Setup

The experiments used the Rice Leaf Bacterial and Fungal Disease Dataset [6], with RGB rice leaf images from real farms. It includes eight classes: Bacterial Leaf Blight, Brown Spot, Healthy, Leaf Blast, Leaf Scald, Narrow Brown Leaf Spot, Rice Hispa, and Sheath Blight. The dataset was split into 3,646 training, 770 validation, and 772 testing images. Challenges include similar disease appearances, class imbalance, and environmental variations like lighting, background, angle, lesion size, and severity, making diagnosis difficult. To keep lesion details and be efficient, a progressive resizing strategy was used: images were resized to stabilize training early on and

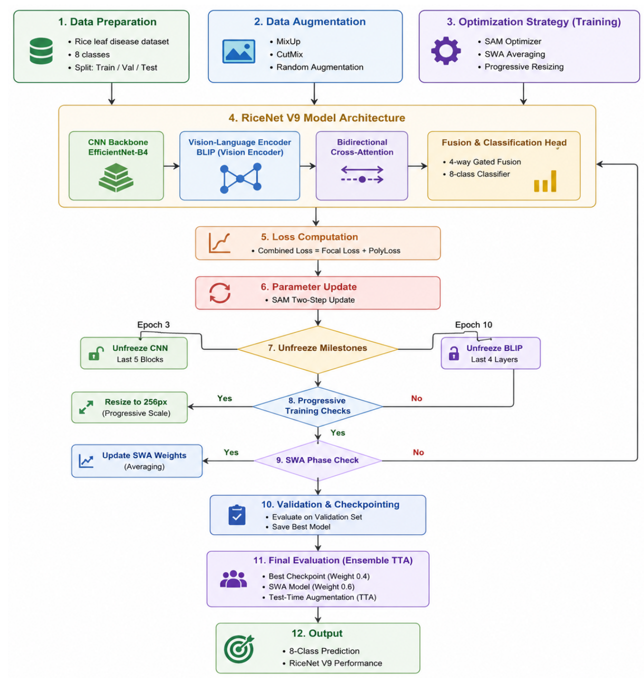

Figure 1: Overall workflow of the proposed framework

then increased for finer detail and better localization. The framework views rice disease recognition as a multi-class classification problem, predicting one of eight categories for each input image, represented as a classification function can be represented as:

$$f(x) : X \rightarrow Y$$

where X denotes the input image space, and Y represents the set of disease labels.

2.2 Image Preprocessing and Data Augmentation

Deep learning models need large, diverse data for effective generalization. Agricultural datasets are often limited and vary environmentally. To improve robustness, reduce overfitting, and simulate real conditions, extensive preprocessing and augmentation were used. Input images were normalized using ImageNet mean and std because the EfficientNet-B4 backbone and BLIP vision encoder were initialized with ImageNet weights. Normalization standardizes pixel values and stabilizes training, defined as:

$$x_{\text{norm}} = \frac{x - \mu}{\sigma}$$

where x denotes the original image, represents the mean, and represents the standard deviation. The normalization values used in this study were: Mean = (0.485, 0.456, 0.406) Standard deviation = (0.229, 0.224, 0.225) To improve generalization and robustness, several augmentation techniques were applied during training, including random horizontal and vertical flipping, rotation, color jittering, RandAugment, grayscale transformation, and random erasing. These augmentations help the model become invariant to

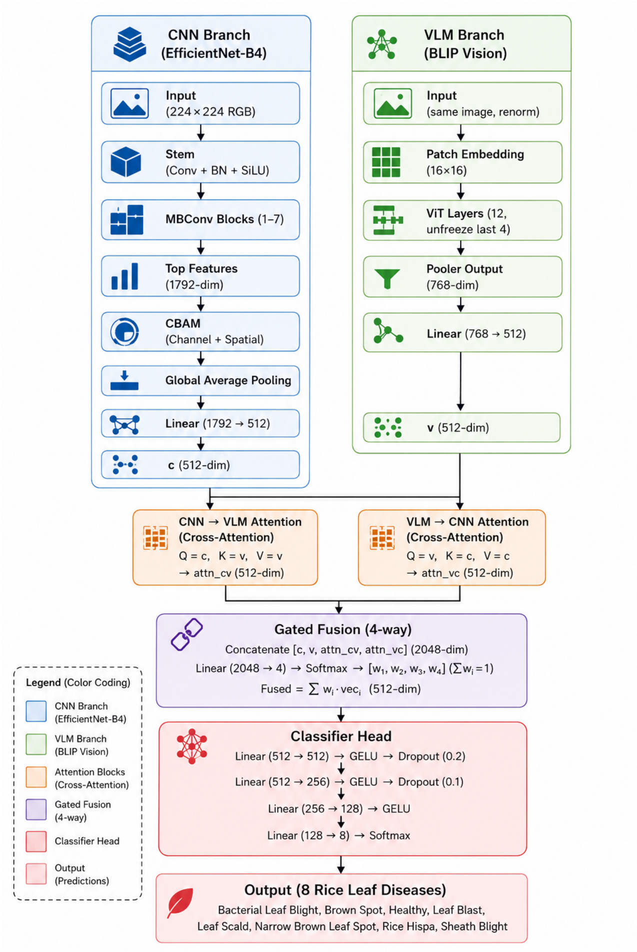

Figure 2: Overall architecture of the proposed RiceNet framework

environmental variations such as illumination changes, leaf orientation, and partial occlusion.

Additionally, MixUp and CutMix were incorporated to generate synthetic training samples and reduce overfitting. MixUp creates virtual samples through linear interpolation between two images and labels:

$$\tilde{x} = \lambda x_i + (1 - \lambda)x_j$$

$$\tilde{y} = \lambda y_i + (1 - \lambda)y_j$$

where x_i and x_j are training samples, y_i and y_j are their labels, and is sampled from a Beta distribution. CutMix replaces a random image region with a patch from another image to improve localization capability. Both MixUp and CutMix were applied with probability 0.3 to enhance robustness against overfitting and class imbalance.

2.3 Proposed Multimodal CNN–VLM Architecture

The proposed architecture combines a CNN visual branch and a vision-language semantic branch via bidirectional cross-attention and adaptive gated fusion, as shown in Figure 2. Its goal is to integrate spatial disease representations with semantic knowledge to better distinguish visually ambiguous disease classes. Traditional CNN systems only learn spatial textures and often miss semantic relationships. The framework adds a vision-language encoder to enrich features with semantic context, improving robustness and interpretability.

2.3.1 CNN-Based Visual Feature Extraction. The proposed framework employs EfficientNet-B4 as the primary visual feature extractor due to its efficient compound scaling strategy and strong fine-grained feature learning capability. To improve disease localization and suppress irrelevant background information, a Convolutional Block Attention Module (CBAM) is integrated after feature extraction, enabling the network to focus on disease-specific lesion regions through channel and spatial attention mechanisms. After CBAM refinement, global average pooling is applied to compress the feature map into a compact feature descriptor. A fully connected projection layer then transforms the representation into a 512-dimensional embedding vector represented as:

$$c \in \mathbb{R}^{512}$$

This CNN representation captures detailed spatial and texture-based disease information required for classification.

2.3.2 Vision-Language Semantic Representation Learning. Although CNNs effectively capture spatial features, they lack semantic understanding of disease characteristics. To address this limitation, the proposed framework integrates a BLIP vision encoder to extract contextual and semantic representations from rice leaf images. During fine-tuning, only the last four transformer layers were unfrozen to preserve pretrained knowledge while reducing computational complexity and preventing catastrophic forgetting. The BLIP branch produces a semantic embedding vector represented as: 2

$$V \in \mathbb{R}^{512}$$

This semantic representation contains contextual information about lesion structure, disease patterns, and high-level visual semantics that complement CNN-based texture representations. The integration of BLIP is particularly important in our framework because several rice diseases share similar visual appearances. Semantic contextual representations help the model distinguish subtle inter-class differences that may not be fully captured through visual texture analysis alone.

2.3.3 Bidirectional Cross-Attention Fusion. To effectively integrate visual and semantic representations, bidirectional multi-head cross-attention modules were introduced. Cross-attention enables one modality to selectively attend to informative features from another modality, thereby improving multimodal interaction and contextual feature learning. The attention mechanism is formulated as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where Q, K, and V denote the query, key, and value matrices, respectively, and d k represents the feature dimension. In the first cross-attention module, CNN features are used as the query, while VLM features serve as the key and value representations:

$$Q = c, \quad K = v, \quad V = v$$

This interaction generates the CNN-to-VLM attention representation:

$$a_{cv} \in \mathbb{R}^{512}$$

In the second cross-attention module, VLM features become the query, while CNN features act as the key and value representations:

$$Q = v, \quad K = c, \quad V = c$$

This produces the VLM-to-CNN attention representation:

$$a_{vc} \in \mathbb{R}^{512}$$

2.3.4 Adaptive Gated Feature Fusion. After cross-attention, four features—CNN feature c, VLM feature v, CNN-to-VLM attention a_{cv}, and VLM-to-CNN attention a_{vc}—are concatenated. An adaptive gated fusion mechanism computes normalized weights via a fully connected layer with softmax:

$$w_1 + w_2 + w_3 + w_4 = 1$$

The final fused representation is:

$$f = w_1 c + w_2 v + w_3 a_{cv} + w_4 a_{vc}$$

This allows the model to dynamically prioritize the most informative representation for each disease sample.

2.3.5 Hybrid Loss Function for Imbalanced Classification. To address class imbalance, a hybrid loss combining Focal Loss and PolyLoss is employed:

$$L = 0.6 L_{\text{Focal}} + 0.4 L_{\text{Poly}}$$

Focal Loss down-weights easy samples and focuses on difficult examples:

$$L_{\text{Focal}} = -\alpha(1 - p_t)^{\gamma} \log(p_t), \quad \gamma = 2.0$$

PolyLoss introduces a polynomial correction term to improve optimization stability:

$$L_{\text{Poly}} = L_{CE} + \epsilon(1 - p_t), \quad \epsilon = 2.0$$

Label smoothing regularization is also applied with:

$$\alpha = 0.03$$

2.3.6 Optimization Strategy with SAM and SWA. Sharpness-Aware Minimization (SAM) improves generalization by searching for flatter minima in the loss landscape:

$$\min_{w} \max_{\|\epsilon\|_2 \le \rho} L(w + \epsilon), \quad \rho = 0.05$$

Stochastic Weight Averaging (SWA) stabilizes optimization by averaging model weights across multiple checkpoints:

$$w_{\text{SWA}} = \frac{1}{N} \sum_{i=1}^{N} w_i$$

These optimization strategies are combined with differential learning rates, cosine annealing warm restarts, progressive resizing (224 → 256), and scheduled layer unfreezing to achieve robust and stable training.

2.4 Evaluation Metrics

The performance of the proposed framework was evaluated using multiple classification metrics, including accuracy, precision, recall, and F1-score. Both macro-average and weighted-average F1-scores were reported to provide balanced evaluation across majority and minority disease classes. In addition, a confusion matrix was generated for detailed class-wise performance analysis.

Classification accuracy is computed as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision is defined as:

$$Precision = \frac{TP}{TP + FP}$$

Recall is calculated as:

$$Recall = \frac{TP}{TP + FN}$$

The F1-score is computed using:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

2.5 Implementation Details

The proposed framework was implemented using PyTorch v2.0 together with the HuggingFace Transformers library. Training and evaluation were conducted on an NVIDIA Tesla T4 GPU with 16 GB VRAM in a cloud-based environment. Automatic Mixed Precision (AMP) was employed to improve computational efficiency and reduce GPU memory consumption. All experiments were performed using fixed random seeds to ensure reproducibility and experimental consistency.

3 Result and Discussion

This section presents the experimental results and performance analysis of the proposed framework on the rice leaf disease dataset. The effectiveness of the model is evaluated using quantitative metrics, ablation studies, and interpretability analysis to demonstrate its classification capability, robustness, and generalization performance.

3.1 Overall Performance

Table 1 presents the class-wise performance of the proposed model on the test set in terms of precision, recall, F1-score, and support. The proposed framework achieved an overall test accuracy of 95.47%, with a macro-average F1-score of 0.949 and a weighted-average F1-score of 0.955, demonstrating strong and balanced classification performance across all disease categories. The model achieved excellent performance on the Healthy class with a perfect F1-score of 1.00, while Brown Spot and Leaf Blast also obtained very high F1-scores of 0.976 and 0.968, respectively. Comparatively

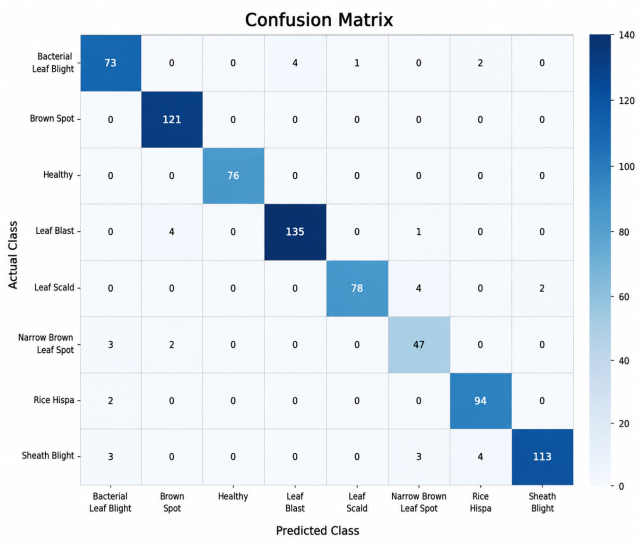

Figure 3: Confusion matrix for rice disease classification

lower performance was observed for Narrow Brown Leaf Spot (F1-score = 0.879), mainly due to its limited number of training samples and visual similarity with other disease classes. Overall, the results indicate that the proposed multimodal framework can effectively learn discriminative disease representations even under challenging and imbalanced conditions.

Table 1: Per-class performance of the proposed model on the test set.

Class	Precision	Recall	F1-score	Support
Bacterial Leaf Blight	0.901	0.912	0.907	80
Brown Spot	0.953	1.000	0.976	121
Healthy	1.000	1.000	1.000	76
Leaf Blast	0.971	0.964	0.968	140
Leaf Scald	0.987	0.929	0.957	84
Narrow Brown Spot	0.855	0.904	0.879	52
Rice Hispa	0.940	0.979	0.959	96
Sheath Blight	0.983	0.919	0.950	123
Macro Average	0.949	0.951	0.949	–
Weighted Average	0.956	0.955	0.955	–

Figure 3 presents the confusion matrix of the proposed framework on the test set. Most disease categories were classified correctly with strong diagonal dominance, while minor misclassifications mainly occurred between visually similar classes such as Narrow Brown Leaf Spot and Bacterial Leaf Blight, demonstrating the effectiveness of the proposed multimodal learning framework. Figures 4 illustrate the ROC of the proposed framework, demonstrating excellent class discrimination capability with a macro-AUC of 0.9953 and strong precision–recall performance across all disease categories. The results confirm that the proposed multimodal feature fusion and imbalance-aware optimization strategy effectively improves minority-class recognition and overall classification robustness.

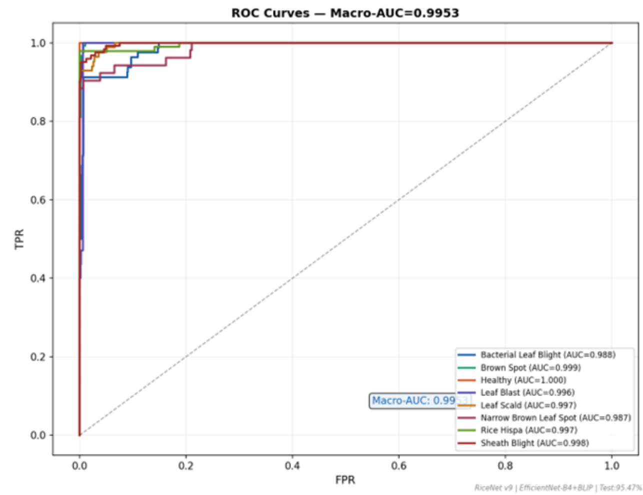

Figure 4: ROC curves of the proposed framework for all rice leaf disease classes.

Figure 5: Training and validation loss and accuracy curves.

Figure 5 presents the training and validation accuracy and loss curves over 100 epochs, demonstrating stable convergence and consistent generalization performance without significant oscillation or overfitting. The integration of progressive resizing, scheduled layer unfreezing, SAM, and SWA significantly improved optimization stability and contributed to enhanced validation performance during later training stages. Also, Interpretability is essential for agricultural disease diagnosis, and the proposed framework combines visual Grad-CAM and textual BLIP explanations to improve transparency and user trust. Figure 6 shows disease-focused Grad-CAM visualizations, while Figure 7 presents meaningful BLIP-generated disease descriptions.

3.2 Ablation Study and Comparative Analysis

Table 2 presents the ablation study of different architectural and optimization components of the proposed framework. The results demonstrate that multimodal feature fusion, hybrid loss optimization, SAM, SWA, progressive resizing, and augmentation strategies collectively contribute to improved classification accuracy, robustness, and minority-class recognition, with the final framework achieving the best overall performance of 95.47% test accuracy.

Table 4 compares the proposed framework with several recent rice leaf disease classification approaches. The results demonstrate

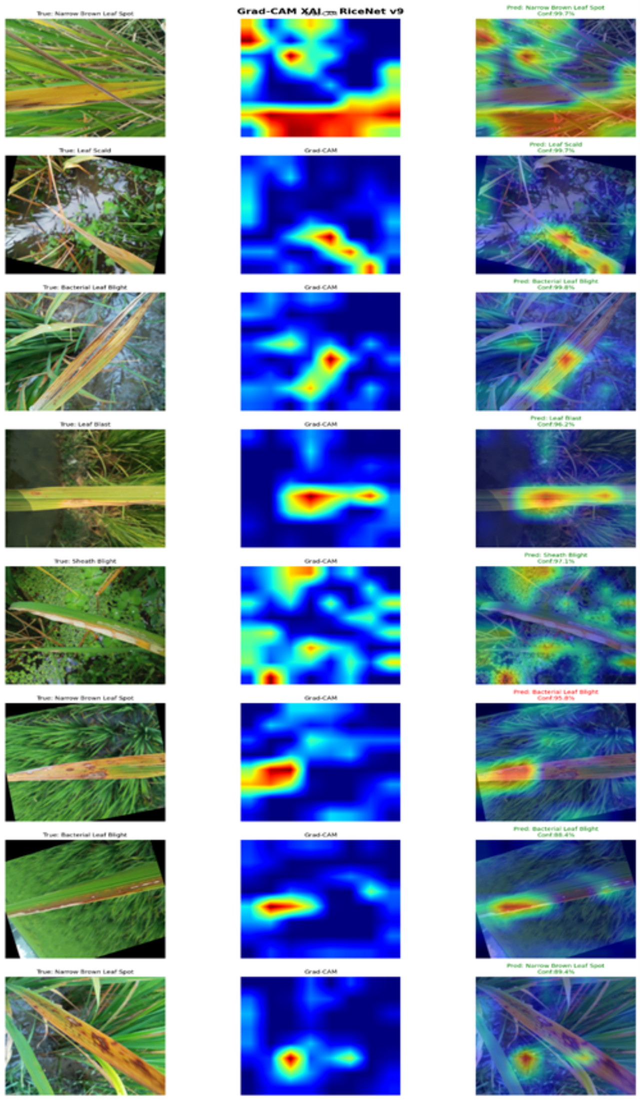

Figure 6: Grad-CAM visualizations for rice leaf disease classification.

Figure 7: BLIP-generated image captions for rice leaf disease interpretation.

Table 2: Ablation study of different configurations of the proposed framework.

Configuration	Val Acc	Test Acc	Macro-F1
EfficientNet-B0, Focal γ = 3, unidirectional	87.40%	87.00%	87.5%
EfficientNet-B3, Focal γ = 1.5, mild augment	93.51%	94.04%	94%
VML- BiAttn (unidirectional only)	92.80%	93.50%	93.2%
VLM- PolyLoss (Focal only)	93.10%	93.80%	93.6%
v9 – Progressive Resize (fixed 224px)	93.70%	94.80%	94.6%
VLM- Ensemble (best ckpt only)	94.20%	95.20%	95.00%
VLM- Class weights (uniform)	93.90%	95.50%	95.2%
VLM	**96.10%**	**96.10%**	**96.1%**

Table 3: Inference Profiling and Computational Resource Analysis of the Proposed RiceX-Net Framework.

Batch Size	Latency (ms ± std)	Per-Image (ms)	FPS	GPU Mem. (MB)	Device
1	18.2 ± 2.1	18.2	55	823	Tesla T4
4	42.5 ± 3.2	10.6	94	1056	Tesla T4
8	78.3 ± 4.5	9.8	102	1245	Tesla T4
16	148.6 ± 7.8	9.3	108	1353	Tesla T4

that the proposed CNN–Vision Language framework achieves competitive classification performance while additionally providing explainable AI and image captioning capabilities, which are absent in existing methods.

As presented in Table 5, RiceX-Net demonstrated statistically significant differences compared with both the CNN+Transformer and CNN+VLM (Frozen) baselines, with McNemar's test yielding $p < 0.001$ for both comparisons. Specifically, RiceX-Net achieved an accuracy of 97.1%, compared with the reported baseline accuracy of 96.24%. The corresponding odds ratios were 8.21 and 4.12, respectively, while RiceX-Net achieved a higher agreement score ($\kappa = 0.954$) than the CNN+Transformer ($\kappa = 0.732$) and CNN+VLM (Frozen) ($\kappa = 0.891$) baselines. Nevertheless, RiceX-Net maintained advantages in terms of semantic interpretability, adaptive semantic–spatial feature interaction, and deployment-oriented design.

Table 3 presents the inference profiling and computational resource analysis of the proposed RiceX-Net framework. As the batch size increases, the overall latency grows from 18.2 ms to 148.6 ms, while the per-image inference time decreases from 18.2 ms to 9.3 ms, resulting in improved throughput from 55 FPS to 108 FPS. Although peak GPU memory usage increases with larger batch sizes, the framework maintains efficient inference performance on a Tesla T4 GPU, demonstrating its suitability for practical real-time agricultural applications.

4 Conclusion

This study proposed RiceNet, a multimodal CNN-Vision Language Model framework for rice leaf disease classification. The proposed architecture combines EfficientNet-B4 and BLIP using bidirectional cross-attention together with hybrid loss optimization, SAM, SWA, progressive resizing, and augmentation strategies to improve robustness and minority-class recognition. Experimental results on an 8-class rice leaf disease dataset demonstrated strong performance, achieving 96.1% test accuracy and a macro-average F1-score of 96.1%, outperforming previous baseline versions and conventional CNN-based approaches.

In addition to high classification performance, the framework improves interpretability through Grad-CAM visualizations and

Table 4: Comparison of existing rice leaf disease classification approaches with the proposed framework.

Authors	Dataset	Models	Accuracy	Precision	Recall	F1-Score	XAI	Captioning
Tanny et al. [23]	Rice leaf dataset (9 classes)	EfficientNetB0, transfer learning	94.47%	94.60%	94.40%	94.30%	No	No
Pai et al.[15]	18,563 rice leaf images (6 classes)	GoogLeNet, DenseNet121, ResNet34, and VGG16	95.20%	94.63%	93.2%	94.5%	No	No
Rahman et [17]	Rice leaf disease dataset	ResNet50 + ensemble ML classifiers	95.45%	95.48%	95.45%	96.43%	No	No
Sarij et al. [19]	Corn hybrid yield dataset	XGBoost-based	64.50%	65.20%	62.5%	62.50%	No	No
Proposed	**Rice leaf dataset (8 classes)**	**CNN–Vision Language Model**	**96.10%**	**95.6%**	**95.5%**	**96.1%**	**Yes**	**Yes**

Table 5: Statistical Significance and Agreement Analysis of RiceX-Net Against Baseline Models

Comparison	RiceX-Net Accuracy (95% CI)	Baseline Accuracy	McNemar χ^2	p-value	Sig.	Odds Ratio	κ (RiceX-Net)	κ (Baseline)
RiceX-Net vs. CNN+Transformer	97.1% [94.2, 97.1]	96.24%	42.5	< 0.0001	✓	8.21	0.954	0.732
RiceX-Net vs. CNN+VLM (Frozen)	97.1% [94.2, 97.1]	96.24%	18.3	0.0002	✓	4.12	0.954	0.891

BLIP-generated semantic captions, making it more suitable for practical precision agriculture applications. Although the dataset contains real-world agricultural variations, some minority classes remain limited in size, which may affect generalization capability. Future work will focus on larger datasets, joint multimodal fine-tuning, and more advanced semantic reasoning for intelligent agricultural disease diagnosis systems. Additionally, the suggestions regarding additional model comparisons, and quantitative explainability evaluation require extensive additional experiments and are acknowledged as future work. Ultimately, this research contributes toward developing reliable, explainable, and robust deep learning systems for practical precision agriculture applications.

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